Will I be able to cover my monthly expenses?

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Abstract

In this project, three different link functions were evaluated for risk modeling. Model selection techniques revealed that not all the covariates play a role in explainin the phenomena of interest, further examination show that actions were necessary to make inference with the fitted model. After corrections, there were no big differences between the model parameters before and after correction but only in one parameter, further examination showed that no statistically significant evidence of deviations from base levels in some of the categorical covariates exists. Prediction performance was nearly the same for the three link functions and this may suggest the inclusion of more covariates to improve the models in this aspect.

1 Introduction

The access of individuals of all the strata in a population to financial services, also known as financial inclusion, is becoming an important issue in Mexico since it involves the development of public policies that guarantee the protection of individuals but also that are targeted in such a way that financial education is also taken into account, it is because of this that the 2021 National Survey of Financial Inclusion was carried out(Instituto Nacional de Estadística y Geografía, 2021). With this information publicly available there is a wide range of possible data analysis approaches available that will depend upon researchers interests.
In the past years, it has been noticed that Mexican households' debts have increased considerably[1] which, by expert's opinions, has the potential to become a major threaten of the economic system(Tellez, 2022). Even further, recent news papers reports have claimed an increase in the stress that parents of children experience and that are associated to not being able to cover children education-related expenses (Herrera, n.d.), and also there have been independent studies that associate reduced productivity with stress because of debt and monthly expenses(Meza, n.d.).

With the all the above in mind, in the present report the results of modeling the risk of not being able to cover monthly living expenses will be presented. The main objectives of this work are the identification of covariates that can explain the risk of interest but also the development of a model that can be used to predict the risk based on new observations of the set of covariates. To this end, the information of the 2021 Mexican National Survey of Financial Inclusion was used, in particular, the answers to the question *Was your monthly salary enough to cover your expenses?* will be used as the response of interest and will be coupled with a set of covariates my means of appropiate statistical models.

[1]This trend has been noticed since 2012, see (Huerfano, 2012).

The structure of the report is as follows, in section 2.1 asummary of the data will be presented, in section 2.2 the general methodology followed in this work will be presented while in section 2.3 the statistical models will be stated as well as the diagnostic to be used to asses the model assumptions. In section 3.1 the results and interpretations of the different steps of the analysis will be stated, in section 3.2 the results of the model fitted with the selected variables will be show as well as the prediction performance in the training set, and in section 3.3 the performance on unseen data will be reported to select the best model and further prediction will be carried out after the selection of the final model. In section 4 discussion of the results will be presented, and in section 5 the conclusions based on the results will be summarized.

2 Statistical Methods

2.1 Description of the data set

For this work, a subset of the covariates in the original data set from the 2021 Mexican National Survey of Financial Inclusion will be used. This dataset was obtained from the official website of the National Institue of Statistics, Geography and Information[2] and a subset of covariates was selected. In the next table a brief summary of the covariates selected is presented as well as the type of response they belong, i.e, if the corresponding covariate is either, continuous, discrete, categorial, binary, etc.

Table 1: Questions and type of response obtained in the survey that were used in this work.

Question	**Response type**
Was your monthly salary enough to cover your expenses?	Binary
Do you speak any indigenous language?	Binary
Do you have any smartphone?	Binary
Do you keep a record of your income and expenses?	Binary
Do you have any automatic charges to your accounts?	Binary
Have you ever taken a course related to budgeting or responsible use of credits?	Binary
Do you have a savings account?	Binary
Do you have any car?	Binary
Do you have internet access?	Binary
How many rooms does your current home has?	Integer
What is your school level?	Ordinal
What is your marital status?	Nominal
How much is your monthly salary?	Continuous

There are 13554 observations in this dataset and the number of observations in each class of the response of interest are nearly balance, with 6600 ‘No’ answers and 6954 ‘Yes’ answers.

2.2 Modeling strategy

As the first objective of this work is the identification of covariates that could be associated with high or low risk of not covering monthly expenses with with monthly salary, a model

[2]The web site can be access at https://www.inegi.org.mx/programas/enif/2021/

selection step is required and this will be carried out using penalization methods, in particular, the lasso(Tibshirani, 1996) technique will be employed to shrink main effects towards zero in order to identify the important ones among all the covariates in the dataset but it will be coupled with cross-validation via the `glment` package(Tay *et al.*, 2023) for an optimal choice of the hyperparameter λ that controls the ammount of shrinkage. After model selection, inference for the variables already selected will be necessary to asses its impact on the risk and checking for assumptions is an important step to guarantee that inferences made with the model will not be incorrect. In this case, checking assumptions will be conducted using the techniques described in Chapert 8 of (Dunn and Smyth, 2018). As a second objective it is desired to also have a model that predicts well in unobserved cases and predictive performance will be also assessed. However, we would like to test the model with real data and also we should avoid bias as much as possible. Therefore, it was be necessary to split the whole dataset in order to avoid bias in the inference and final model selection steps. To be more precise, the data set was split in three folds, a subset for variable selection, another subset for model fitting and assessment of predictive performance, and a final one that will be also split on other two small datasets for model selection based on predictive performance and another small data set to simulate a real data set for final predictions. Figure 1 below depicts a graphical overview of the different steps involved in this work.

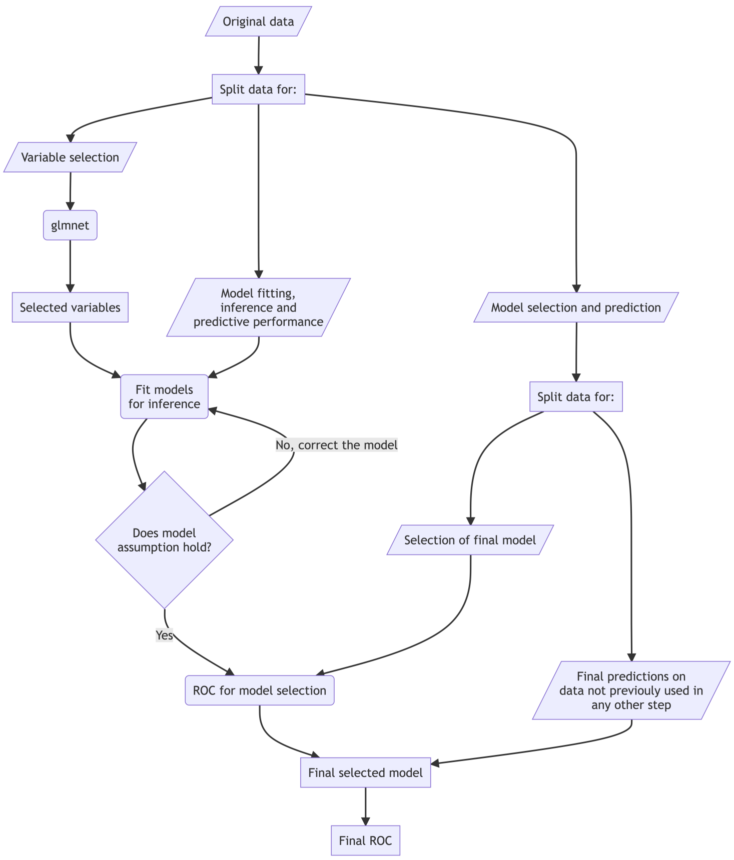

Figure 1: Overview of modeling strategy followed in this work.

2.3 Statistical models

Given the fact that the variable of interest is measured by means of the 'Yes/No' response to a question in a survey, we have to use an appropriate model to handle a binary response. Therefore, a Bernoulli model was used to account for that and then we will follow the Generalized Linear Modeling approach. In the following we will describe and state the models used in both model selection and model fitting for inference steps, and we will make more emphasis on the link functions since for this project the logit link, probit link and complementary log-log link were used in both steps to find out which of them performs better for the data and the objectives. The table below summarizes the link functions:

Table 2: Link functions used and its corresponding functional form.

Link function	**Functional form**
Logit link	$log(p/1-p)$
Probit link	$\Phi^{-1}(p)$
Complementary Log-Log	$log(-log(1-p))$

2.3.1 Model selection step

As mentioned earlier, model selection was carried out using the lasso technique and `glmnet` package. In this step, the linear or systematic component of the model was the same for the three link functions, the corresponding formula is as follows:

$$\begin{aligned}\eta_i = \;& \beta_0 + \beta_1 L_{i,\text{Preschool}} + \beta_2 L_{i,\text{Primary}} + \beta_3 L_{i,\text{Secundary}} + \beta_4 L_{i,\text{Technical-secundary}} \\ & +\beta_5 L_{i,\text{Basic Normal school}} + \beta_6 L_{i,\text{High school}} + \beta_7 L_{i,\text{Technical-high school}} + \beta_8 L_{i,\text{Bachelors}} \\ & +\beta_9 L_{i,\text{Graduate}} + \beta_{10} M_{i,\text{Not married}} + \beta_{11} M_{i,\text{Married}} + \beta_{12} M_{i,\text{Apart}} \\ & +\beta_{13} M_{i,\text{Divorced}} + \beta_{14} M_{i,\text{Widow}} + \beta_{15} X_{i,\text{Indigenous}} + \beta_{16} X_{i,\text{Salary}} \\ & +\beta_{17} X_{i,\text{Smartphone}} + \beta_{18} X_{i,\text{Expenses}} + \beta_{19} X_{i,\text{Charges}} + \beta_{20} X_{i,\text{Courses}} \\ & +\beta_{21} X_{i,\text{Savings}} + \beta_{22} X_{i,\text{Cards}} + \beta_{23} X_{i,\text{Bedrooms}} + \beta_{24} X_{i,\text{Bathrooms}} \\ & +\beta_{25} X_{i,\text{Car}} + \beta_{26} X_{i,\text{Internet}}\end{aligned} \tag{1}$$

Where $L_{i,\text{j}}$ represents an indicator variable for category j of School level covariate in the i-th surveyed person, $M_{i,\text{k}}$ are the indicator variables for category k in Marital status covariate in the i-th surveyed person, $X_{i,\text{Salary}}$ is the ammount of monthly salary, $X_{i,\text{Smartphone}}$ is an indicator variable for whether the i-th surveyed person has a smartphone or not, $X_{i,\text{Expenses}}$ is an indicator variable for whether the person surveyed keeps a record of incomes and expeses, $X_{i,\text{Charges}}$ is an indicator variable for whether the person has automatic charges to their accounts, $X_{i,\text{Courses}}$ is an indicator for wheter or not the person has ever taken a course in financial education and responsible use of credit, $X_{i,\text{Savings}}$ is an indicator for whether the person has or not a savings account, $X_{i,\text{Cards}}$ is a covariate for the number of credit cards the person has, $X_{i,\text{Bedrooms}}$ is a covariate for the number of bedrooms in the current house of the person, $X_{i,\text{Bathrooms}}$ is a covariate for the number of bathrooms in the current house of the person, $X_{i,\text{Car}}$ is an indicator of whether the person has a car or nor and finally, $X_{i,\text{Internet}}$ is an indicator for whether or not the person has internet access in his/her house.

The rationale for using the description in the equation above is that we are interested in finding the covariates that could aid in the explanation of the phenomena under study and we can think of it as a screening experiment rather than an in depth study of the interactions between the covariates. Then, the models for variable selection can be written as follows:

- $y_i \sim Binomial(1, p(x_i))$; $logit(p(x_i)) = \eta_i = \beta x^T$
- $y_i \sim Binomial(1, p(x_i))$; $probit(p(x_i)) = \eta_i = \beta x^T$
- $y_i \sim Binomial(1, p(x_i))$; $cloglog(p(x_i)) = \eta_i = \beta x^T$

Where cloglog stands for the complementary log log link.

2.3.2 Model fitting for inference

In this step, all the parameters that are non zero after the lasso variable selection step will be included in the model. However, it may happen that indicator variables associated to one of the multicategory covariates will have almost all their parameters equal to zero. In that case, if at least one of the parameters associated is non zero the main effect will be included in the next step. For assesing the assumptions of the model the diagnostics strategies described in (Dunn and Smyth, 2018) were followed, and to be more specific Quantile residuals were computed to assess the adequacy of the systematic component but also to check for the appropriateness of the binomial distribution as a random component. To check for outliers, absolute value of Studentized residuals were computed and plotted to look for those observations above 2 or 3, and finally Cook's distance was used to check for influential observations.

3 Analysis and results

3.1 Model selection results

After that model selection was carried out using `glmnet`, the results were examined and the only non zero parameters were those associated to the intercep β_0, $L_{i,\text{Technical-secundary}}$, $L_{i,\text{Bachelors}}$, $X_{i,\text{Indigenous}}$, $X_{i,\text{Smartphone}}$, $X_{i,\text{Expenses}}$, $X_{i,\text{Charges}}$, $X_{i,\text{Savings}}$, $X_{i,\text{Bathrooms}}$, $X_{i,\text{Car}}$, $X_{i,\text{Internet}}$. Then, the final formula for the systematic component, following the guidelines of the previous section, is as follows:

$$
\begin{aligned}
\eta_i = \;& \beta_0 + \beta_1 L_{i,\text{Preschool}} + \beta_2 L_{i,\text{Primary}} + \beta_3 L_{i,\text{Secundary}} + \beta_4 L_{i,\text{Technical-secundary}} \\
& + \beta_5 L_{i,\text{Basic Normal school}} + \beta_6 L_{i,\text{High school}} + \beta_7 L_{i,\text{Technical-high school}} + \beta_8 L_{i,\text{Bachelors}} \\
& + \beta_9 L_{i,\text{Graduate}} + \beta_{10} X_{i,\text{Indigenous}} + \beta_{11} X_{i,\text{Smartphone}} + \beta_{12} X_{i,\text{Expenses}} \\
& + \beta_{13} X_{i,\text{Charges}} + \beta_{14} X_{i,\text{Savings}} + \beta_{15} X_{i,\text{Bathrooms}} + \beta_{16} X_{i,\text{Car}} \\
& + \beta_{17} X_{i,\text{Internet}}
\end{aligned} \tag{2}
$$

3.2 Model fitting and prediction

After the variables were selected, the models for estimation and inference on the paremeters were fitted using the formula in (2). The results concerning null and residual deviance, and AIC are presented in the table below.

Table 3: Null deviance, residual deviance and AIC values for the three models with different link functions.

Model's link function	Null Deviance	Residual deviance
Logit	6261.2 on 4517 df	6029.6 on 4500 df
Probit	6261.2 on 4517 df	6030.3 on 4500 df
CLogLog	6261.2 on 4517 df	6034.9 on 4500 df

Figures 2 to 5 show diagnostic plots for each of the models and the components.

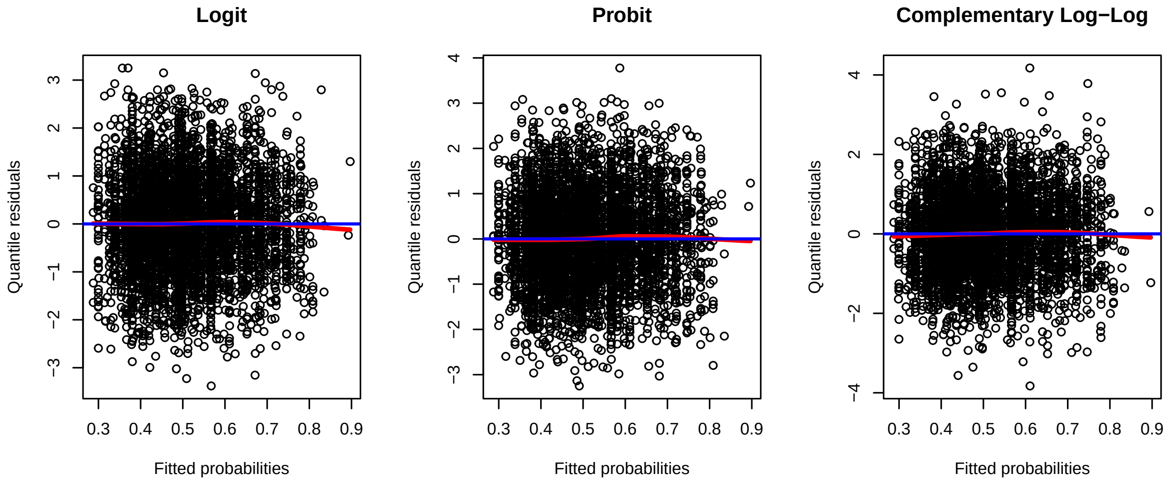

Figure 2: Quantile residuals for the systematic component. No trend indicates adequacy of the corresponding link function. The blue line represents the constant function at y = 0, the red line is a smoothing line.

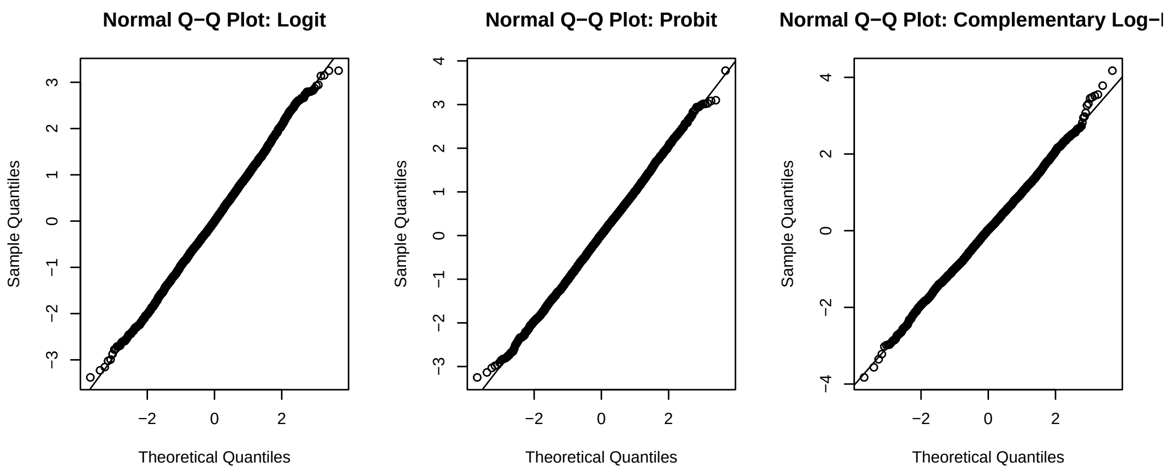

Figure 3: Normal quantile plots for quantile residuals of the fitted models. Deviations from normality in the plots indicate problems with the chosen distribution.

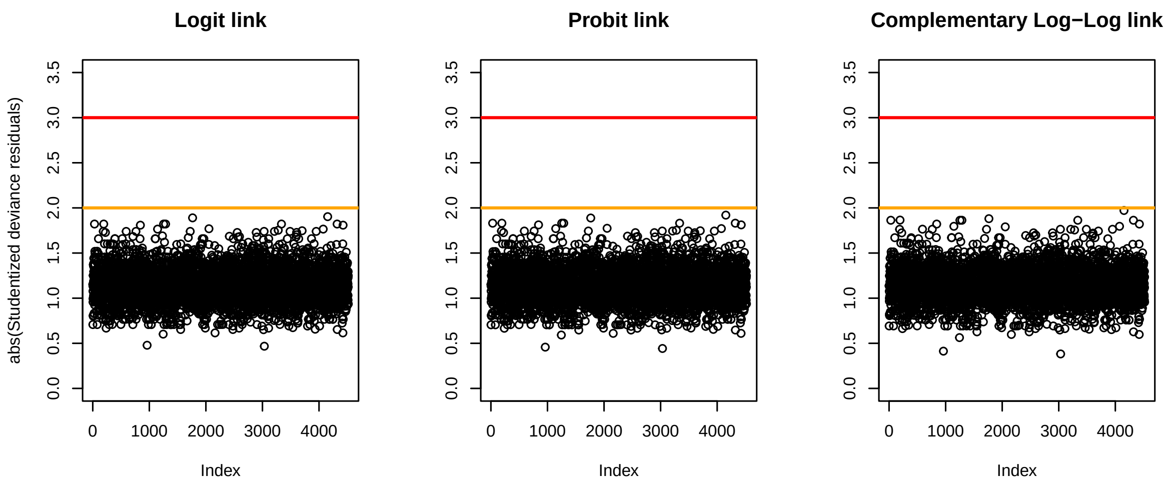

Figure 4: Rstudentized residuals for each model. Values greater that 2 may be a flag for warning.

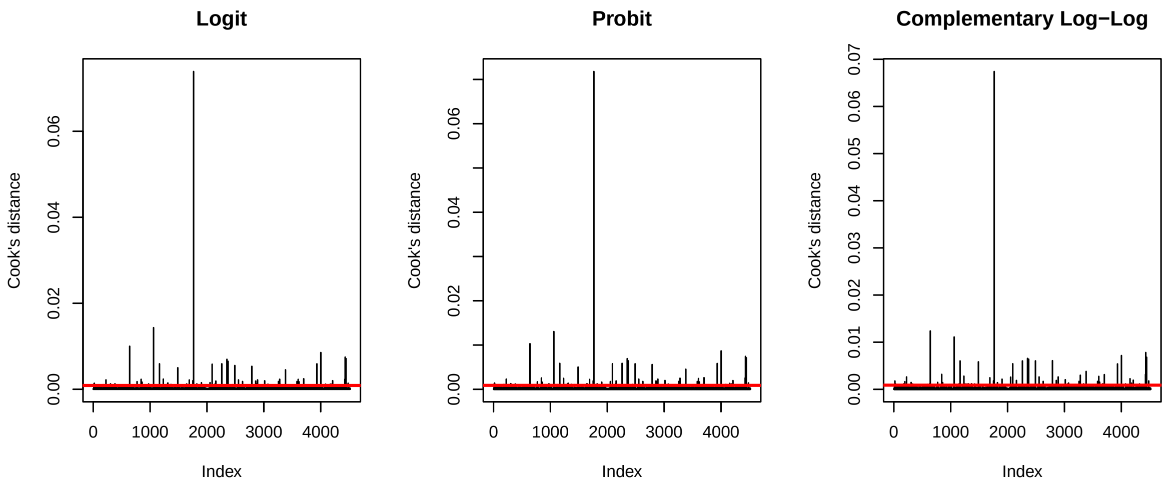

Figure 5: Cook's distance for each model. Threshold set at 4/n. There are various potential influential points but one of them is clearly a sign of warning.

Table 4: Change in the parameters estimated after removing potential influential points in each model.

	Logit		Probit		CLogLog	
	Before	After	Before	After	Before	After
Intercept	-0.532	-0.530	-0.328	-0.327	-0.749	-0.747
school_levelPreschool	1.512	12.930	0.938	4.882	1.022	3.314
school_levelPrimary	-0.137	-0.138	-0.086	-0.086	-0.095	-0.095
school_levelSecundary	-0.069	-0.069	-0.043	-0.043	-0.034	-0.034
school_levelTechnical-secundary	-0.143	-0.143	-0.087	-0.086	-0.062	-0.062
school_levelBasic Normal school	-0.102	-0.102	-0.064	-0.064	-0.103	-0.103
school_levelHigh school	0.208	0.208	0.129	0.129	0.152	0.153
school_levelTechnical-high school	-0.132	-0.131	-0.080	-0.080	-0.069	-0.069
school_levelBachelors	0.210	0.211	0.131	0.131	0.153	0.153
school_levelGraduate	0.344	0.345	0.206	0.207	0.204	0.205
speak_indigenouslangYes	-0.179	-0.185	-0.112	-0.115	-0.138	-0.143

any_smartphoneYes	0.163	0.161	0.101	0.100	0.127	0.125
earning_expenses_historyYes	0.127	0.127	0.077	0.077	0.074	0.074
automatic_monthly_chargesYes	0.321	0.321	0.195	0.196	0.197	0.197
savings_accountYes	0.362	0.360	0.222	0.221	0.231	0.230
num_bathrooms	0.181	0.180	0.110	0.110	0.108	0.108
any_carYes	0.305	0.305	0.190	0.190	0.214	0.214
internet_accessYes	-0.058	-0.058	-0.035	-0.035	-0.027	-0.027

As can be seen in the dignostics plots, only a possible influential point was found and after correcting for it the model were fitted again and there was no big change in the parameters as can be see in Table 4 but all model agree that the paramter associated to School level Preschool was the most affected. At the end of the analysis only the covariates associated to having a smartphone, having automatic monthly charges, having a savings account, the number of bathrooms in the house and if the person has a car or not were found to be non zero at 95% confidence, all of them with positive coefficients. Only interpretations in the probit are available in the fact that all the coefficients were positive indicate that, marginally, each of categorical predictos has more risk when the answer to the question in the survey is ‘Yes’ and also, as the number of bathrooms increases so does the risk.

3.3 The best model for Mexican population

To summarize the predictive performance, Area Under the Curve plots was computed and displayed in Figure 6. As can be see in Table 5 below, there is practically no difference between the three models in terms of prediction performance. Therefore any of the models could be used.

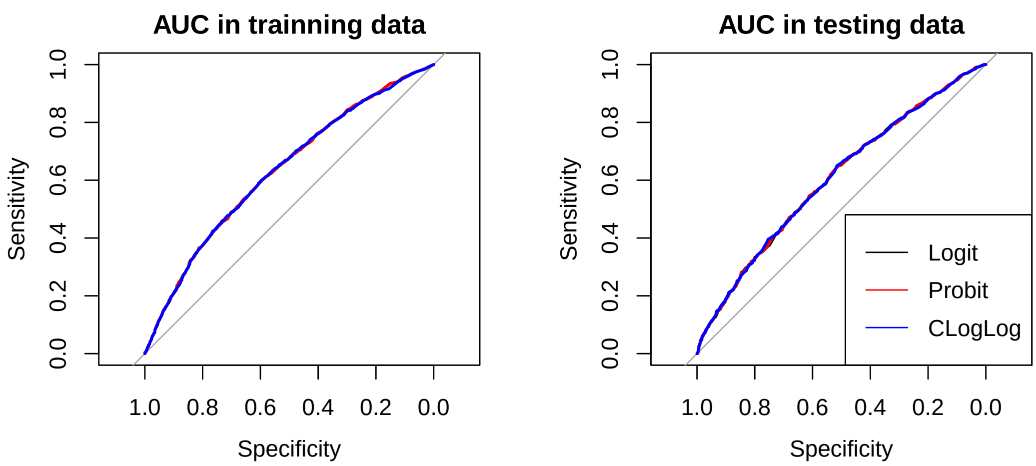

Figure 6: Area under the curve plots for the three models in both the trainning set and the testing set.

Table 5: Area under the curve for the three models in both trainning and testing sets.

	AUC fit data	AUC unobserved data
Logit	0.6315440	0.6052297
Probit	0.6315440	0.6054118

CLogLog	0.6309448	0.6054715

4 Discussion

The fact that model selection indicate that salary was not an important covariate is surprising but at the same time it is interesting, this behaviour may indicate that the dynamics of debt goes beyond only salary or the amount an individual earn in a monthly basis. Further, martial status was not present in the covariates selected. Also, the number of credit cards did not show up as a valuable predictor, and this may have happend because not a lot of people have credit cards. The three link functions seem to be appropriate for both inference and prediction tasks, however, the fact that other link functions than logit link do not allow interpretations make it difficult to choose them in practice. Also, given that in prediction tasks the three link functions appear not to have any difference, then an easy to interpret model would be desirable. In terms of model adequacy, all diagnostic plots indicate that the only concern was the existence of an influential point, and after correcting for it there was not a big difference but just in one parameter in the three models. Regarding predictive performance, all the model behave almost the same in this task, therefore any of the models could be used for future predictions. However the large values of the residual deviance indicate that the models fitted in this project do not fit as good as the saturated model.

5 Conclusions and recommendations

Three link functions were tested for prediction of the risk of not covering monthly expenses with monthly salary. Model selection selection step show that marital status, salary, to have taken courses in financial education, the number of credit cards nor the number of bedrooms in the house of mexican people play a role in explaining the phenomena of interest. Further model fitting and inference revealed that there exist no statistically significant deviations of models parameters from that of the base level of no school at all in the school level covariate, the same was observed for whether or not a person speaks an indigenous language, whether or not people keep a record of their incomes and expenses and for whether or not people have internet access at home. The three models tested perform quite similar for prediction and the inclusion either of higher order interactions and more covariates could improve the predictions. Also, other approaches may work better, for example desicion trees, since most of the covariates are factors with different levels.

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Appendix: R code used to generate the report

```
library(here); library(tidyverse); library(viridis); library(glmnet)
data.dir <- here('data', 'cleandata')
#Loading the data
enif2021_subset <- read_csv(here(data.dir, 'risk_assesment_nsfi2021.csv'),
                            show_col_types = F)
enif2021_subset <- enif2021_subset %>%
  mutate(across(where(is.character), as.factor))
enif2021_subset <-
  enif2021_subset %>%
  mutate(
    school_level =
      fct_relevel(school_level,
                  c('No school', 'Preschool','Primary','Secundary',
                    'Technical-secundary','Basic Normal school',
                    'High school', 'Technical-high school',
                    'Bachelors', 'Graduate')),
    marital_status = fct_relevel(marital_status,
                  c('Single' , 'Not married', 'Married',
                    'Apart', 'Divorced', 'Widow'))
  )
#Splitting the data in K-folds (K = 18)
K_folds <- 3; set.seed(2023-13-11) #A seed for reproducibility
fold_id <-
  #Create a vector that indicates the fold number for each sample
  rep(1:K_folds, each = nrow(enif2021_subset)/K_folds) %>%
  #Compute a random permutation of the vector that contains
  #the fold number for each sample. In this way we try to
  #avoid any particular clustering of the data in a given fold
  sample()
enif2021_subset <-
  enif2021_subset %>% mutate('fold_class' = as.factor(fold_id))
#Splitting the data in three small data sets for the
#different analysis to be performed:
#a) Data for model selection
model_selection.data <- enif2021_subset %>% filter(fold_class==1) %>%
  select(!c('FOLIO', 'fold_class'))
#b) Data for model fitting and inference
fitting_inference.data <- enif2021_subset %>% filter(fold_class==2) %>%
  select(!c('FOLIO', 'fold_class'))
#c) Data for model prediction and performance assessment
model_assesment.data <- enif2021_subset %>% filter(fold_class==3) %>%
  select(!c('FOLIO', 'fold_class'))
#We will fit a first model that includes only all the main effects
model1_formula <- earned_enough ~ .
model1.matrix <- model.matrix(model1_formula, data = model_selection.data)
model1.matrix <- model1.matrix[,-1] #Removing intercept for glmnet
#glmnet variable selection
```

```r
set.seed(2023-16-12) #A seed for reproducibility
variable_selection.canonical <-
  cv.glmnet(
    y = model_selection.data$earned_enough, x = model1.matrix,
    family = 'binomial', nfolds = 18)
coefs.canonical <- coef(variable_selection.canonical)
set.seed(2023-16-12) #A seed for reproducibility
variable_selection.probit <-
  cv.glmnet(
    y = model_selection.data$earned_enough, x = model1.matrix,
    family = binomial(link = 'probit'), nfolds = 18)
coefs.probit <- coef(variable_selection.probit)
set.seed(2023-16-12) #A seed for reproducibility
variable_selection.cloglog <-
  cv.glmnet(
    y = model_selection.data$earned_enough, x = model1.matrix,
    family = binomial(link = 'cloglog'), nfolds = 18
  )
coefs.cloglog <- coef(variable_selection.cloglog)
cbind(
  rownames(coefs.canonical)[as.matrix(coefs.canonical) != 0],
  rownames(coefs.probit)[as.matrix(coefs.probit) != 0],
  rownames(coefs.cloglog)[as.matrix(coefs.cloglog) != 0]
)

#Formulas for selected models
model_canonical <- earned_enough ~ school_level+ speak_indigenouslang +
  any_smartphone + earning_expenses_history +  automatic_monthly_charges +
  savings_account + num_bathrooms + any_car + internet_access
model_probit <- earned_enough ~ school_level+ speak_indigenouslang +
  any_smartphone + earning_expenses_history +  automatic_monthly_charges +
  savings_account + num_bathrooms + any_car + internet_access
model_cloglog <- earned_enough ~ school_level+ speak_indigenouslang +
  any_smartphone + earning_expenses_history +  automatic_monthly_charges +
  savings_account + num_bathrooms + any_car + internet_access
##### Model fitting steps
inference_canonical <- glm(model_canonical, family = binomial(),
                           data = fitting_inference.data)
inference_probit <- glm(model_probit, family = binomial(link = 'probit'),
                        data = fitting_inference.data)
inference_cloglog <- glm(model_cloglog, family = binomial(link = 'cloglog'),
                         data = fitting_inference.data)
inference_canonical %>% summary(): inference_probit %>% summary()
inference_cloglog %>% summary()
######### Model diagnostics
#Checking the systematic component
qresids_canonical <- statmod::qresid(inference_canonical)
qresids_probit <- statmod::qresid(inference_probit)
qresids_cloglog <- statmod::qresid(inference_cloglog)
```

```r
par(mfrow = c(1,3))
scatter.smooth(qresids_canonical ~ fitted(inference_canonical),
               ylab = 'Quantile residuals', xlab = 'Fitted probabilities',
               main = 'Logit link', lpars = list(col = 'red', lwd = 3))
abline(h = 0, col = 'blue', lwd = 2)
scatter.smooth(qresids_probit ~ fitted(inference_canonical),
               ylab = 'Quantile residuals', xlab = 'Fitted probabilities',
               main = 'Probit link', lpars = list(col = 'red', lwd = 3))
abline(h = 0, col = 'blue', lwd = 2)
scatter.smooth(qresids_cloglog ~ fitted(inference_canonical),
               ylab = 'Quantile residuals', xlab = 'Fitted probabilities',
               main = 'Complementary Log-Log link',
               lpars = list(col = 'red', lwd = 3))
abline(h = 0, col = 'blue', lwd = 2); par(mfrow = c(1,1))
#Cheking the random component
par(mfrow = c(1,3))
qqnorm(qresids_canonical, main = 'Normal Q-Q Plot: Logit link')
qqline(qresids_canonical)
qqnorm(qresids_probit, main = 'Normal Q-Q Plot: Probit link')
qqline(qresids_probit)
qqnorm(qresids_cloglog, main = 'Normal Q-Q Plot: Complementary Log-Log link')
qqline(qresids_cloglog)
par(mfrow = c(1,1))
#Cheking for outliers
rstud_canonical <- rstudent(inference_canonical)
rstud_probit <- rstudent(inference_probit)
rstud_cloglog <- rstudent(inference_cloglog)
par(mfrow = c(1,3))
plot(abs(rstud_canonical),  ylab = 'abs(Studentized deviance residuals)',
     main = 'Logit link', ylim = c(0, 3.5),)
abline(h = 2, col = 'orange', lwd = 2); abline(h = 3, col = 'red', lwd = 2)
plot(abs(rstud_probit),  ylab = '', main = 'Probit link', ylim = c(0, 3.5),)
abline(h = 2, col = 'orange', lwd = 2); abline(h = 3, col = 'red', lwd = 2)
plot(abs(rstud_cloglog), ylab = '', main = 'Complementary Log-Log link',
     ylim = c(0, 3.5))
abline(h = 2, col = 'orange', lwd = 2); abline(h = 3, col = 'red', lwd = 2)
par(mfrow = c(1,1))
#Cheking for influential onbservations
cooksD_canonical <- cooks.distance(inference_canonical)
cooksD_probit <- cooks.distance(inference_probit)
cooksD_cloglog <- cooks.distance(inference_cloglog)
flag_val <- 4/nrow(fitting_inference.data)
par(mfrow = c(1,3))
plot(cooksD_canonical, type = 'h', main = 'Logit link',
     ylab = "Cook's distance")
abline(h = flag_val, col = 'red', lwd = 2)
plot(cooksD_probit, type = 'h',
     main = 'Probit link', ylab = "Cook's distance")
abline(h = flag_val, col = 'red', lwd = 2)
```

```
plot(cooksD_cloglog, type = 'h',
     main = 'Complementary Log-Log link', ylab = "Cook's distance")
abline(h = flag_val, col = 'red', lwd = 2); par(mfrow = c(1,1))
#Removing potential influential point to see what happens
inference_canonical.2 <- update(inference_canonical,
                                subset = (-which.max(cooksD_canonical)))
inference_probit.2 <- update(inference_probit,
                             subset = (-which.max(cooksD_probit)))
inference_cloglog.2 <- update(inference_cloglog,
                              subset = (-which.max(cooksD_cloglog)))
#Comparing the coefficients to see if the results change
cbind(inference_canonical$coefficients,inference_canonical.2$coefficients)
cbind(inference_probit$coefficients, inference_probit.2$coefficients)
cbind(inference_cloglog$coefficients, inference_cloglog.2$coefficients)
# ROC and AUC for trainning data
logit_ROC <- pROC::roc(inference_canonical.2$model$earned_enough ~
                         inference_canonical.2$fitted.values)
probit_ROC <- pROC::roc(inference_probit.2$model$earned_enough ~
                          inference_canonical.2$fitted.values)
cloglog_ROC <- pROC::roc(inference_cloglog.2$model$earned_enough ~
                           inference_cloglog.2$fitted.values)
logit_ROC %>% plot( xlim = c(1,0), ylim = c(0,1))
probit_ROC %>% plot(add = T, col = 'red')
cloglog_ROC %>% plot(add = T, col = 'blue')
#Assesing predictive performance
mask.assesment <- sample(1:nrow(model_assesment.data),
                         size = nrow(model_assesment.data)/2)
model_assesment.data1 <- model_assesment.data[mask.assesment,]
model_assesment.data2 <- model_assesment.data[-mask.assesment,]
#ROC for selecting the final model
logit.assesment_predictions <- predict(inference_canonical.2,
                                       newdata = model_assesment.data1,
                                       type = 'response')
probit.assesment_predictions <- predict(inference_probit.2,
                                        newdata = model_assesment.data1,
                                        type = 'response')
cloglog.assesment_predictions <- predict(inference_cloglog.2,
                                         newdata = model_assesment.data1,
                                         type = 'response')
logit_ROC.assesment <- pROC::roc(model_assesment.data1$earned_enough ~
                         logit.assesment_predictions)
probit_ROC.assesment <- pROC::roc(model_assesment.data1$earned_enough ~
                          probit.assesment_predictions)
cloglog_ROC.assesment <- pROC::roc(model_assesment.data1$earned_enough ~
                           cloglog.assesment_predictions)
auc_byModel <-
  data.frame(
    'Model' = c('Logit', 'Probit', 'CLogLog'),
    'ROC_fit' = c(logit_ROC$auc, probit_ROC$auc, cloglog_ROC$auc),
```

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    'ROC_assesment' = c(logit_ROC.assesment$auc, probit_ROC.assesment$auc,
                        cloglog_ROC.assesment$auc)
  )
#Final predictions
logit.final_preds <- predict(inference_canonical.2,
                                    newdata = model_assesment.data2,
                                    type = 'response')
logit.finalROC <- pROC::roc(model_assesment.data2$earned_enough ~
                              logit.final_preds)
probit.final_preds <- predict(inference_probit.2,
                              newdata = model_assesment.data2,
                              type = 'response')
probit.finalROC <- pROC::roc(model_assesment.data2$earned_enough ~
                               probit.final_preds)
cloglog.final_preds <- predict(inference_cloglog.2,
                               newdata = model_assesment.data2,
                               type = 'response')
cloglog.finalROC <- pROC::roc(model_assesment.data2$earned_enough ~
                                cloglog.final_preds)
auc_byModel$'Real_auc' <- c( logit.finalROC$auc, probit.finalROC$auc,
                             cloglog.finalROC$auc)
auc_byModel
```